

The Intelligent Gambler: An Artificial Intelligence Model for Optimal Play and Betting in Blackjack

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Abstract—Card games, as well as casino games as a whole, make it incredibly difficult for the players at hand, due to the unpredictable nature of their ultimately random environments. You could have every strategy memorized, for any hand imaginable, yet winning is not guaranteed. Blackjack is a particularly special case, as there are statistically verifiable strategies for beating casinos and making a profit. The motivation for this project is to study the effectiveness of these strategies and to reveal characteristics of potentially near-perfect play in a curated environment. We posture to combine previously studied and meticulously optimized AI strategies with "card counting", to implement an AI model that can statistically win in a standard casino environment.

I. INTRODUCTION

With Blackjack, the nature of the game is to hope for the statistically best possible set of cards and beat the dealer with your actions based on these cards. The objective is clear, but the uncertainty of the obstacles presents the biggest problem. When the biggest independent variable being the cards dealt is outside of your control, you lose agency. Blackjack is a unique game in that sense, as in most games your environment can be somewhat controlled by your actions, in Blackjack you cannot influence what cards come to the hand. This makes knowledge of the environment the most powerful tool a player can wield, and is why the casino industries are always thriving, as they are always the ones in control of the environment.

Our project aims at implementing an AI model in Blackjack that mirrors the motivations previously mentioned, where the model can infer the game-play dynamically to produce an optimal strategy through the concept of "counting cards", the ability to keep track of the cards that have been dealt in the past to

forecast the cards being dealt in the future. The goal of our project is for our AI model to optimize our potential for winnings in Blackjack by inferring the circumstances of an ongoing Blackjack game through its analysis of the environment by dictating the best action dependent on one's hand and the cards that have already been played, and providing the best course of action to reach a win. Thus as opposed to simply reacting turn-by-turn, the agent is adaptive to the changing environment to administer the best action.

For clarity, the rules of *Blackjack*:

- Each player, including the dealer, is initially dealt two cards.
- Players either choose to hit, stand, double, or split.
- Players hit until they choose to stand or until their total exceeds 21, resulting in a bust, and they lose the hand. Dealers must stand on 17.
- When you double down you are allowed to double your bet after receiving your first 2 cards, then only dealt one after.
- If you are dealt a pair (two cards of the same rank) you can split these into two separate hands. At which point you must make an additional bet equal to your starting bet. The dealer will give you a second card to go with the first split card. You will then decide to hit or stand. After you play out this hand and stand, you will move on to the next split card.
- If both the dealer and player have the same hand number, a push occurs and the player receives their bet back.

The objective of the game is to have a hand that beats the dealer's hand by reaching a total card value

of 21 or as close to 21 as possible.

With basic strategy, your chances of winning in blackjack can be as high as 42.22%, with a 8.48% chance of a tie (push). The probability of losing is approximately 49.10%.

This shows an obvious advantage that the casino holds over the player. But with card counting, a player can garner odds as large as a 2% advantage, effectively beating the dealer and winning money in the long-term.

Agent-based modeling is particularly important due to the characteristics of Blackjack. Evidently, in the game, you have some sensors in the cards dealt, and actions that a player can choose to perform based on these sensors. As well, past environments or actions may affect a decision in the future due to the strategy of "card counting". These characteristics make Agent-based modeling an obvious abstraction for this type of problem, due to the existence of a dynamically changing environment, being the cards dealt, and an agent which acts upon the environment, the player.

In Blackjack, the sole agent involved is the player. The dealer is not considered an agent, because they follow a strict algorithm that is void of any decision-making. The player's primary goal is to beat the dealer, through decisions dictated by the state of the game, their hand of cards, and the potential of the dealer's hand. In other words, the player's goal is to beat the environment. The player infers the cards that have been played as a whole, takes into consideration the value of their hand, and makes a judgment on which of several possible actions they should make.

The environment is the game setting that involves the player and the dealer, the cards dealt and the cards held. Also, an environment's past state has an effect on an environment's future state, which means that the player must remember previous turns to optimize their chances of winning. A player interacts with the environment by iteratively and continuously choosing from a set of finite and constant decisions each turn, whereby the environment may react based on which action was made. More cards could be dealt, the dealt cards could reset; the player could win, or the player could lose.

II. RELATED WORK

Artificial intelligence has experienced significant advancements across various domains, including the realm of card games. In this context, Blackjack stands out as a game where the player's disadvantage against the house is notably lower compared to other casino offerings. A research paper, "Application of Neuroevolution in Blackjack" [1] utilized basic strategy charts, establishing a player's theoretical win rate at approximately 42.6%. The study then applied the NEAT (Neuroevolution of Augmenting Topologies) algorithm, an evolutionary algorithm that creates artificial neural networks, to develop an AI-driven strategy for Blackjack that manages to marginally surpass the basic strategy with a win rate of 42.68%, an increase of 0.08%. Despite this improvement, the dealer's win rate in simulations involving the NEAT-based strategy was also higher at 49.54% (push accounting for 7.9%) compared to the basic strategy. This discrepancy is attributed to the AI's focus on strategy development without integrating card counting techniques, an aspect of Blackjack that could potentially enhance decision-making and increase the odds to a player advantage.

A public repository was discovered, "Blackjack-Bot" [2], which utilizes reinforced learning and Q-learning, techniques aimed at teaching an AI model through iterative improvements. They successfully simulated a Blackjack environment with an AI player, but there is still room for improvement. The inherent disadvantage against the casino remains a limitation, as even with the optimal strategy derived by the AI, the casino still has a slight edge over the player. To extend this work, we are exploring avenues to address this inherent disadvantage, focusing on strategic decision-making enhancements. Our goal is to provide the AI player with tools for making dynamic and informed bets, aiming to tilt the odds in its favor. This effort, combined with the AI's derived optimal strategy, holds the potential to turn the game profitable.

The original project's primary focus on Q-learning and the development of Basic Strategy successfully reduced the casino's edge to just 0.5%, laying the foundation for our proposed extension. Our contribu-

	2	3	4	5	6	7	8	9	T	A
20	S	S	S	S	S	S	S	S	S	S
19	S	S	S	S	S	S	S	S	S	S
18	S	S	S	S	S	S	S	S	S	S
17	S	S	S	S	S	S	S	S	S	S
16	S	S	S	S	S	S	H	H	H	H
15	S	S	S	S	S	S	H	H	H	H
14	S	S	S	S	S	S	H	H	H	H
13	S	S	S	S	S	S	H	H	H	H
12	H	H	S	S	S	S	H	H	H	H
11	D	D	D	D	D	D	D	D	D	D
10	D	D	D	D	D	D	D	D	D	D
9	H	D	D	D	D	D	H	H	H	H
8	H	H	H	H	H	H	H	H	H	H
7	H	H	H	H	H	H	H	H	H	H
6	H	H	H	H	H	H	H	H	H	H
5	H	H	H	H	H	H	H	H	H	H

Classic Baseline Blackjack Strategy

	2	3	4	5	6	7	8	9	T	A
A-9	S	S	S	S	S	S	S	S	S	S
A-8	S	S	S	S	D	S	S	S	S	S
A-7	D	D	D	D	D	S	S	H	H	H
A-6	H	D	D	D	D	H	H	H	H	H
A-5	H	H	D	D	D	H	H	H	H	H
A-4	H	H	D	D	D	H	H	H	H	H
A-3	H	H	H	D	D	H	H	H	H	H
A-2	H	H	H	D	D	H	H	H	H	H

	2	3	4	5	6	7	8	9	T	A
A-A	P	P	P	P	P	P	P	P	P	P
T-1	S	S	S	S	S	S	S	S	S	S
9-9	P	P	P	P	P	P	P	P	P	P
8-8	P	P	P	P	P	P	P	P	P	P
7-7	P	P	P	P	P	P	P	H	H	H
6-6	P	P	P	P	P	P	H	H	H	H
5-5	D	D	D	D	D	D	D	D	H	H
4-4	H	H	H	P	P	H	H	H	H	H
3-3	P	P	P	P	P	P	H	H	H	H
2-2	P	P	P	P	P	P	H	H	H	H

Fig. 1. Optimal Strategy for Blackjack (H = hit, S = stand, D = double, P = split), from [1]

tion will involve dynamic and strategic betting using a well-known Blackjack technique to give our AI a more nuanced approach than the project’s original one. This strategic approach will aim to address the inherent limitations and seek a more effective way to gain an edge over the casino.

In a public article tailored towards data science, “Winning Blackjack using Machine Learning” [3] explores how a genetic algorithm (GA), inspired by evolutionary principles, can be used to craft an effective strategy for playing Blackjack. The article emphasizes that due to the house edge, an optimal strategy’s primary aim is to minimize losses rather than guaranteeing consistent wins. While the GA can generate and refine strategies across generations, it alone cannot provide the AI player with a sufficient advantage to outsmart casinos.

The article compares the strategy generated by the GA to a recognized optimal solution for Blackjack, showcasing the effectiveness of the technique. It demonstrates the potential of genetic algorithms in solving complex problems with large solution spaces such as the game of Blackjack. The article further discusses the role of fitness functions, which measure a strategy’s success, and highlights the importance of playing a large number of hands to evaluate strategies accurately. Various configurations, including population size, selection methods, mutation rates, and termination conditions, are explored to fine-tune the GA’s perfor-

mance.

It’s crucial to recognize the inherent limitations of the current approach. Despite AI’s ability to create and refine strategies, they all still succumb to the same flaws as the originally derived optimal strategy—the casino’s edge over the player. Our proposed extension seeks to address the inherent limitations and challenges associated with these approaches. One of the key objectives is to develop strategies that go beyond minimizing losses and actively work towards gaining a strategic advantage over the casino. By focusing on refining decision-making strategies and reducing the reliance on chance to turn a profit, we aim to create an AI player that can navigate the complexities of Blackjack more effectively, ultimately mitigating the impact of the house’s advantage.

The “Monte Carlo Blackjack” Project [4], focused on applying reinforcement learning, specifically Monte Carlo Tree Search (MCTS), to Blackjack. The research framed the game as a Partially Observable Markov Decision Process (POMDP), seeking optimal policies for different game states. The POMDP MCTS approach integrates previous actions and observations to influence the expected value of a state and action, employing algorithms to simulate and update values in a Monte Carlo Tree Search framework for Partially Observable Markov Decision Processes. This methodology involves multiple simulations and adjustments based on simulated outcomes. Despite its insights, the study had limitations such as excluding many aspects of Blackjack like splitting pairs, doubling-down, and reshuffling, as well as assuming a minimal impact of game sequences on future rounds.

To build upon this work, we plan to introduce a more comprehensive set of rules and abilities that encompass all the complexities of the game. This includes incorporating more advanced decision-making strategies and expanding our scope to handle a broader set of Blackjack variants. A new contribution we plan on implementing involves incorporating dynamic betting into our own AI Blackjack player project. While the existing work focused on estimating optimal policies based on observed states, our addition will introduce a

new strategic element where the AI will have to think about the state of the game and most importantly, the deck, to make a "good" bet. This enhancement will force the model to track the composition of the deck over multiple rounds, in turn enabling more sophisticated decision-making and potentially outperforming traditional strategies.

Inherent in all these works is a shared limitation—they struggle to turn a profit consistently. This challenge stems from the persistent advantage held by the house. Even with a flawless blackjack strategy, the dealer maintains a 0.5% chance of winning over the player. Recognizing this hurdle, a well-known Blackjack strategy named card counting was devised to counteract the house edge. Our project stands out as we plan to integrate the card counting strategy as well as dynamic betting into our Agent's toolkit. Card counting, a technique involving the tracking of card values, allows our bot to gain a significant advantage by adjusting its strategy and bet based on the composition of the deck. Dynamic betting will allow our Agent to adjust its bets higher or lower when it deems the odds are more or less favorable. By implementing these strategies in tandem, our AI seeks to not only overcome the limitations observed in existing works but also markedly improve its performance, potentially securing consistent winnings against the casino. The potential strength of this enhancement will lie in the Agent's ability to synergize the established optimal strategies with card counting, to provide dynamic insights for maximizing the AI player's advantage.

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